

1 0. Prof's Blueprint (High Priority)

- Errors:** abs/rel error, MAE, RMSE on discrete points.
- Polynomial forms:** standard vs nested (Horner) vs centered; FLOP counts.
- Horner:** evaluate $p(c)$; synthetic division by $ax - b$; derivative via repeated Horner.
- Taylor:** construct poly at given center; truncation error integral & Lagrange form; bound on interval (Sandwich Thm); find n for tolerance; Maclaurin/Taylor series convergence.
- Interpolation:** setup linear system + matrix form $Xa = y$; Vandermonde det (why nonzero); over/underdetermined & free parameters; Lagrange basis $\ell_i(x)$ & poly $p(x)$; extrapolation vs interpolation; divided differences (1st/2nd order only, no full Newton); Chebyshev nodes, Runge phenomenon; Bézier curves (basic).
- MCQ + written section, 30 pts total. Bring 1 A4 cheat sheet (this one!).

2 1. Error Formulas

$$E_{abs} = |f(x) - \tilde{f}(x)|, \quad E_{rel} = \frac{|f(x) - \tilde{f}(x)|}{|f(x)|}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |f(x_i) - \tilde{f}(x_i)|$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (f(x_i) - \tilde{f}(x_i))^2}$$

Trap: RMSE penalizes large errors more (squares); MAE treats all equally. Round to 5 dp unless told otherwise.

3 2. Polynomial Forms

Standard: $p_n(x) = a_n x^n + \dots + a_1 x + a_0$ — needs $\sim 2n - 1$ mults, n adds (naive) or $n(n+1)/2$ mults w/o precomputed powers.

Nested (Horner): $p_n(x) = (\dots((a_n x + a_{n-1})x + a_{n-2})x + \dots)x + a_0$ — n mults, n adds. **Always fewer ops.**

Centered at x_0 : $p_n(x) = \sum b_k (x - x_0)^k$. Get b_k via repeated synthetic division by $(x - x_0)$, or Taylor: $b_k = p^{(k)}(x_0)/k!$.

Horner's Method (synthetic division)

Divide $p(x)$ by $(x - c)$: bring down a_n , each next coeff = $a_k + c \cdot$ (prev result). Last value = remainder = $p(c)$.

$$\begin{array}{c|cccc} & a_n & a_{n-1} & \cdots & a_0 \\ c & & cb_{n-1} & \cdots & cb_0 \\ \hline & b_{n-1} & b_{n-2} & \cdots & r \end{array}$$

- $p(x) = (x - c)q(x) + r$, $r = p(c)$; bottom row (minus last) = coeffs of $q(x)$.
- Divide by $(ax - b)$: divide by $(x - b/a)$ first (root $c = b/a$), get $q^*(x)$; true quotient $q(x) = q^*(x)/a$, same remainder r .
- Derivative link:** run the tableau on $p(x)$ to get $q(x)$ & $r = p(c)$; run *another* tableau on $q(x)$ at same $c \Rightarrow$ new remainder = $p'(c)$. (Repeat on next quotient for $p''(c)/2!$, etc.).
- Trap:** two ways to get $p'(1)$ — direct Horner on $p'(x)$, or double-tableau on $p(x)$ — must match.

4 3. Taylor & Maclaurin

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x)$$

Maclaurin: $x_0 = 0$.

Truncation Error $R_n(x)$

Integral form:

$$R_n(x) = \int_{x_0}^x \frac{(x-t)^n}{n!} f^{(n+1)}(t) dt$$

Lagrange form:

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$$

for some ξ between x_0 and x . **Relationship:** Lagrange form = Mean Value Thm applied to the integral form (weighted avg of $f^{(n+1)}$ over interval).

Bounding $R_n(x)$ on interval $[a, b]$ (Sandwich Thm)

- Find $M = \max_{\xi \in [a, b]} |f^{(n+1)}(\xi)|$ (often = bound on derivative amplitude, e.g. for $\sin, \cos$ terms $M \leq 3^{n+1}$ etc.).

- Find max of $|x - x_0|^{n+1}$ on $[a, b]$ (largest distance from center).

$$3. |R_n(x)| \leq \frac{M \cdot d^{n+1}}{(n+1)!} \text{ where } d = \max |x - x_0|.$$

Finding n for tolerance $|R_n| < \varepsilon$

Set bound expression $< \varepsilon$, solve for smallest integer n (try increasing n until factorial growth beats d^{n+1} growth). Use $n!$ growth dominates exponential — trial/error with a table is acceptable.

"Largest x " type (Quiz 3 Q5 style)

Fix n ; treat bound $\frac{M d^{n+1}}{(n+1)!} < \varepsilon$ as inequality in x (since $d = x - x_0$ here); solve for x .

Series Convergence

Taylor series = $f(x) \iff \lim_{n \rightarrow \infty} R_n(x) = 0$. Writing the infinite sum from the pattern does *not* by itself prove convergence — must separately verify $R_n \rightarrow 0$ (true for $\sin, \cos, e^x$ on all $\mathbb{R}$ since M bounded, $d^{n+1}/(n+1)! \rightarrow 0$).

Maclaurin of $\sin(3x)$

$$\begin{aligned} \sin(3x) &= \sum_{k=0}^{\infty} \frac{(-1)^k (3x)^{2k+1}}{(2k+1)!} \\ &= 3x - \frac{27x^3}{6} + \frac{243x^5}{120} - \dots \end{aligned}$$

Trap: degree-8 Maclaurin poly of an odd function still stops at x^7 term (even-degree coeffs are 0) — write poly *through* degree 8, don't force an x^8 term.

5 4. Chain Rule / Derivative Practice

$[f(g(x))]' = f'(g(x))g'(x)$. Common combos: $2 \sin x + e^x$; $1 + \cos^2 x = 1 + (\cos x)^2 \Rightarrow$ derivative = $-\sin x \sin x = -\sin(2x)$. **Trap:** evaluate derivatives *at the center* x_0 before plugging into Taylor coeff $f^{(k)}(x_0)/k!$ — don't leave symbolic. Useful values: $\sin(\pi/2) = 1, \cos(\pi/2) = 0$; $\sin(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$; $\sin(-\pi/4) = -1/\sqrt{2}$.

6 5. Interpolation Basics

Interpolation: passes exactly through given data points (local fit). **Approximation:** fits overall trend, need not pass through points (global fit, e.g. least squares). **Extrapolation:** evaluate poly *outside* node range — much less reliable than interpolation (points 4-5 pattern in quizzes: always flag extrapolation as less trustworthy).

Degree-1 (Linear) Interpolation

Given $(x_0, y_0), (x_1, y_1)$: $p_1(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$.

Vandermonde System

For $n+1$ points, poly degree n : $a_0 + a_1 x_i + \dots + a_n x_i^n = y_i$ for each i . Matrix form $Xa = y$ where

$$X = \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{pmatrix}$$

$$\det(X) = \prod_{0 \leq i < j \leq n} (x_j - x_i)$$

Why nonzero: interpolation nodes x_i are *distinct* $\Rightarrow$ every factor $(x_j - x_i) \neq 0 \Rightarrow \det \neq 0 \Rightarrow$ unique solution guaranteed.

Free Parameters (over/underdetermined)

points = $m+1$, poly degree = n . If $n > m$: **underdetermined** — infinitely many interpolating polys; free parameters = $n - m$ (extra coefficients not pinned down by the $m+1$ equations). If $n < m$: **overdetermined** — generally no exact solution (unless data is consistent) $\rightarrow$ use approximation (least squares) instead. **Trap:** "degree at most n " with only $m+1 < n+1$ points $\Rightarrow$ family has $n-m$ free parameters — identify them explicitly as the highest-order coefficients.

Lagrange Interpolation

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}, \quad p(x) = \sum_{i=0}^n y_i \ell_i(x)$$

Properties: $\ell_i(x_j) = \delta_{ij}$ (1 if $i = j$, else 0); basis *depends only on nodes* x_i , not on y_i or f — same ℓ_i reused for any function on same nodes. **Steps:** (1) list nodes; (2) build each $\ell_i(x)$ as product of $(x - x_j)/(x_i - x_j)$ over $j \neq i$; (3) sum $y_i \ell_i(x)$; (4) simplify/evaluate at target point; (5) compute abs/rel error vs known true value if given; (6) extrapolate = plug x outside node range into same $p(x)$.

7 6. Divided Differences (basic)

$$\text{1st order: } f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$$

$$\text{2nd order: } f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

(No need for full Newton formula per prof — just compute these building blocks.)

8 7. Grids, Runge, Chebyshev, Bézier

- Uniform grid:** equally spaced nodes. **Non-uniform:** unequal spacing.
- Runge's phenomenon:** high-degree interpolation on *uniform* nodes over $[-1, 1]$ -type intervals can oscillate wildly near endpoints as n increases.
- Chebyshev nodes:** cluster near interval endpoints; specifically chosen to *minimize* Runge oscillation — preferred over uniform for high-degree interpolation.
- Bézier curves:** defined by control points; curve passes through *endpoint* control points only (not interior ones); used in computer graphics for smooth curve design; based on Bernstein polynomials as blending functions.

9 8. Worked-Pattern Checklists

Error Question Checklist

□ compute $f(x_i)$ exactly □ compute $\tilde{f}(x_i)$ □ abs error = $|diff|$ □ rel error = $abs/|f|$ □ if multiple points: MAE (avg of abs), RMSE (sqrt avg of sq).

Horner Question Checklist

□ write coeffs $a_n, \dots, a_0$ (**include zeros for missing powers!**) □ build synthetic division table (bring down, multiply by c , add) □ last entry = $p(c)$ □ row above = coeffs of $q(x)$ □ repeat table on $q(x)$ at same c for $p'(c)$ if asked □ for divisor $ax - b$: use $c = b/a$, remember to divide quotient coeffs by a .

Taylor Construction Checklist

□ pick center x_0 □ compute derivatives $f, f', f'', \dots$ at x_0 up to needed order □ divide each by $k!$ □ assemble $\sum \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$ □ do *not* simplify away the $(x-x_0)^k$ factors unless asked to expand.

Truncation Error Bound Checklist

□ identify $f^{(n+1)}(x)$ pattern (for sin, cos combos it's bounded by amplitude) □ find $M = \max|f^{(n+1)}|$ on given interval □ find $\max|x-x_0| = d$ on interval □ plug into $Md^{n+1}/(n+1)!$ □ for "find n ": increase n until bound $< \varepsilon$, show smallest such n □ for "find x ": solve bound $< \varepsilon$ for x algebraically or numerically.

Interpolation System Checklist

□ count points $m+1$ vs desired degree n □ write $m+1$ equations $a_0+a_1x_i+\dots+a_nx_i^n=y_i$ □ arrange as $Xa=y$ (rows=points, cols=powers $0..n$) □ if square: compute Vandermonde det via product formula □ if not square: state under/overdetermined & # free params.

Lagrange Checklist

□ list all nodes □ for each needed ℓ_i : multiply $(x-x_j)/(x_i-x_j)$ over all $j \neq i$ (don't forget *every* other node, incl. non-adjacent ones) □ sum $y_i\ell_i(x)$ for full $p(x)$ □ evaluate at target, compare to true value for errors □ extrapolate = same formula outside range, note reduced reliability.

10 9. Common Traps Summary

- Forgetting **zero coefficients** for missing powers in Horner tables.
- Confusing **quotient** $q(x)$ degree (always one less than $p(x)$) after synthetic division.
- Sign errors in $(x-x_0)$ vs (x_0-x) in centered form / Taylor.
- Odd/even function Maclaurin series — half the coefficients vanish; don't invent nonzero terms.
- Mixing up **integral** vs **Lagrange** form of R_n — integral needs no unknown ξ , Lagrange does.
- Free parameter count: it's (poly degree +1) – (# data points), not degree itself.
- Lagrange basis $\ell_i(x_j) = 0$ for $j \neq i$, = 1 for $j = i$ — sanity check your basis polys.
- Vandermonde det = 0 *only if* two nodes coincide — never zero for distinct interpolation nodes.

- Extrapolation error can be large — always flag lower confidence vs interpolation.
- Degree of sum of two polys = $\max(\deg)$; degree of product = $\text{sum}(\deg)$; watch for **cancellation** making sum degree drop (find α s.t. leading terms cancel).
- FLOP counts: nested form always $\leq$ standard form; state exact counts (n mult + n add for degree n nested).
- Round only final numeric answers to 5 decimals; keep exact fractions/symbols mid-derivation.

11 10. Quick Reference Values

$e \approx 2.71828$, $\pi \approx 3.14159$, $\sqrt{2} \approx 1.41421$, $1/\sqrt{2} \approx 0.70711$. $n!$: $5! = 120$, $6! = 720$, $7! = 5040$, $8! = 40320$, $9! = 362880$, $10! = 3628800$.

12 11. Worked Mini-Examples

Horner: $p(x) = 2x^3 - 3x^2 + 0x + 5$ at $c = 2$
Coeffs: 2, -3, 0, 5.

	2	-3	0	5
2		4	2	4
	2	1	2	9

So $p(2) = 9$; quotient $q(x) = 2x^2 + x + 2$; $p(x) = (x-2)(2x^2 + x + 2) + 9$. **For** $p'(2)$: run another tableau on q 's coeffs 2, 1, 2 at $c = 2$:

	2	1	2
2		4	10
	2	5	12

So $p'(2) = 12$.

Lagrange: nodes (0, 1), (1, 3), (2, 7), **find** $p(1.5)$

$$\ell_0(x) = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2}$$

$$\ell_1(x) = \frac{(x-0)(x-2)}{(1-0)(1-2)} = \frac{x(x-2)}{-1}$$

$$\ell_2(x) = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}$$

$p(x) = 1 \cdot \ell_0(x) + 3 \cdot \ell_1(x) + 7 \cdot \ell_2(x)$. At $x = 1.5$:
 $\ell_0 = \frac{(0.5)(-0.5)}{2} = -0.125$, $\ell_1 = \frac{(1.5)(-0.5)}{-1} = 0.75$,
 $\ell_2 = \frac{(1.5)(0.5)}{2} = 0.375$. $p(1.5) = 1(-0.125) + 3(0.75) + 7(0.375) = -0.125 + 2.25 + 2.625 = 4.75$.

Vandermonde det: nodes 0, 1, 2

$$\det(X) = \prod_{0 \leq i < j \leq 2} (x_j - x_i) \\ = (1-0)(2-0)(2-1) = 1 \cdot 2 \cdot 1 = 2 \neq 0$$

Free Parameters: 2 points, degree ≤ 3

Points (2, 5), (5, -1): $m+1 = 2$, so $m = 1$; want degree $n = 3$ (4 coeffs a_0, a_1, a_2, a_3). Equations: 2. Free params = $n - m = 3 - 1 = 2$ — e.g. a_2, a_3 free; solve a_0, a_1 in terms of them from the 2 constraints.

Truncation error bound example

$f(x) = \sin(3x)$, $n = 8$ (so $n+1 = 9$), interval $[-3, 4]$, $x_0 = 0$: $f^{(9)}(x) = \pm 3^9 \cos(3x)$ or $\pm 3^9 \sin(3x) \Rightarrow M = 3^9 = 19683$. $d = \max(|-3|, |4|) = 4$.

$$|R_8(x)| \leq \frac{19683 \cdot 4^9}{9!} = \frac{19683 \cdot 262144}{362880} \approx 14230$$

(Large bound is fine — the point is applying the formula correctly, not that it's tight.)

Diff./Integration Quick Review

Product rule: $(fg)' = f'g + fg'$. Quotient rule:

$$(f/g)' = \frac{f'g - fg'}{g^2}. \quad \text{Chain rule: } [f(g(x))]' =$$

$$f'(g(x))g'(x). \quad \text{Integration by parts: } \int u \, dv =$$

$$uv - \int v \, du. \quad \textbf{Example: } \int x \sin x \, dx: \text{ let } u = x, dv = \sin x \, dx \Rightarrow du = dx, v = -\cos x.$$

$$\int x \sin x \, dx = -x \cos x + \int \cos x \, dx$$

$$= -x \cos x + \sin x + C$$

Centered Form: full worked example

$p(x) = x^3 - 2x^2 + 3x - 5$, center $x_0 = 2$. Repeatedly synthetic-divide by $(x-2)$; each remainder is the next b_k (coefficient of $(x-x_0)^k$).

$$1, -2, 3, -5 \rightarrow 1, 0, 3 \mid b_0=1 \quad (\text{Div.1})$$

$$1, 0, 3 \rightarrow 1, 2 \mid b_1=7 \quad (\text{Div.2})$$

$$1, 2 \rightarrow 1 \mid b_2=4, b_3=1 \quad (\text{Div.3})$$

So $p(x) = 1 + 7(x-2) + 4(x-2)^2 + (x-2)^3$. **Check:** expand back — matches original. □ Each division: bring down leading coeff, next = coeff + 2×prev; last value = remainder.

Horner: divide by $ax - b$ (worked)

Divide $4x^3 - 7x^2 + 5x + 8$ by $2x + 1$ (i.e. $a = 2, b = -1$, root $c = -\frac{1}{2}$).

	4	-7	5	8
$-\frac{1}{2}$		-2	4.5	-4.75
	4	-9	9.5	3.25

$q^*(x) = 4x^2 - 9x + 9.5$, remainder $r = 3.25$ (remainder unchanged by scaling). True quotient $q(x) = q^*(x)/2 = 2x^2 - 4.5x + 4.75$. **Check:** $(2x+1)(2x^2-4.5x+4.75)+3.25 = 4x^3-7x^2+5x+8$. ✓

Bézier Curves: construction

Bernstein form, n control pts $P_0, \dots, P_n$, $t \in [0, 1]$:

$$B(t) = \sum_{i=0}^n \binom{n}{i} (1-t)^{n-i} t^i P_i$$

Quadratic ($n=2$): $B(t) = (1-t)^2 P_0 + 2t(1-t)P_1 + t^2 P_2$. Cubic ($n=3$): add $3t^2(1-t)P_2'$... — use full Bernstein sum above. **Key facts:** curve passes through P_0 (at $t=0$) and P_n (at $t=1$) only; interior points *pull* the curve but aren't touched. **Worked example:** quadratic, $P_0 = (0, 0), P_1 = (1, 2), P_2 = (3, 0)$, find $B(0.5)$:

$$B(0.5) = 0.25(0, 0) + 2(0.5)(0.5)(1, 2) + 0.25(3, 0) \\ = (0, 0) + (0.5, 1) + (0.75, 0) = (1.25, 1)$$